Recap

The activity coefficient of a component in a mixture describes its non-ideal mixing:

$$\mu_i = \left(\frac{\partial G}{\partial n_i}\right)_{T,P,n_j} = \mu_i^{IM} + \mu_i^{ex} = \underline{G}_i + RT \ln x_i + RT \ln \gamma_i$$

The simplest model is the “one-constant Margules equation”:

$$\mu_1^{ex} = RT \ln \gamma_1 = A x_2^{\,2}$$

There are many more complex models available

In all cases the parameters are fit to measurements of activity coefficient

Temperature dependence is achieved via the coefficients of the model

Parameters for binary mixtures can be summed for mixtures with more than two parameters

A simple example of the use of activity coefficient models: Depression of Freezing Point

Consider a pure solid, in equilibrium with a liquid mixture
e.g. ice in sea water

What is the freezing point at a defined liquid composition

Call the component in the solid phase component 1 (water), and the impurity in the liquid component 2 (salt).

Chemical equilibrium condition:

$$\mu_1^S = \mu_1^L \quad \text{at specified } T,P \text{ (1 atmosphere)}$$

Now, $\mu_1^L = \underline{G}_1^L(T,P) + RT\ln\gamma_1 x_1$

ideal mixing term

Pure liquid at same T,P as mixture

non-ideal mixing term

and $\mu_1^S = \underline{G}_1^S(T,P) = \underline{G}_1^L(T,P) - \Delta\underline{G}_1^{fus}(T,P)$

ignore 'cos P-dependence small for liquids/solids

So $RT\ln\gamma_1 x_1 = -\Delta\underline{G}_1^{fus}(T,P)$

equating standard states

But we know that $\underline{G}_1^S = \underline{G}_1^L$ at T_f i.e. $\Delta\underline{G}_1^{fus}(T_f) = 0$

freezing point of pure 1 at atmospheric pressure

and, strictly, 1 atm

We need to relate $\Delta\underline{G}_1^{fus}(T)$ to $\Delta\underline{G}_1^{fus}(T_f)$, which we know

$$d\underline{G} = -\underline{S}dT + \underline{V}dP$$

$$\left(\frac{\partial\Delta\underline{G}_1^{fus}}{\partial T}\right)_P = -\Delta\underline{S}_1^{fus}$$

and so $\Delta \underline{G}_1^{fus}(T) - \Delta \underline{G}_1^{fus}(T_f) = -\Delta \underline{S}_1^{fus}(T - T_f)$

assuming $\Delta \underline{S}_1^{fus}$ is constant over $T \to T_f$

But at T_f ~~and 1 atmosphere,~~ $\Delta \underline{G}_1^{fus} = 0 = \Delta \underline{H}_1^{fus} - T_f \Delta \underline{S}_1^{fus}$

Therefore $\Delta \underline{G}_1^{fus}(T, \sout{1\ \text{atm}}) \approx -\dfrac{\Delta \underline{H}_1^{fus}}{T_f}(T - T_f)$

$$RT \ln \gamma_1 x_1 \approx \frac{\Delta \underline{H}_1^{fus}}{T_f}(T - T_f)$$

$$\boxed{\ln \gamma_1 x_1 \approx \frac{\Delta \underline{H}_1^{fus}}{R}\left(\frac{1}{T_f} - \frac{1}{T}\right)}$$

exact result (almost)

Usually (but not always – see solder question on problem sheet):

$$\Delta T = T_f - T << T_f \text{ and so } T \approx T_f$$

Then: $$\ln \gamma_1 x_1 = \frac{-\Delta \underline{H}_1^{fus}}{R T_f^{\,2}} \Delta T$$

Approximate result for small ΔT

An activity coefficient model gives γ_1 as fn(x_2) for given T

e.g. one constant Margules $RT \ln \gamma_1 = A x_2^{\,2}$

i.e. we have ΔT as fn(x_1)

Note this expression does not include any reference to component 2, i.e. it does not matter what this component is, except in so far as it influences γ_1 (a colligative property)

Also consider what happens if component 2 ionises…..

If component 2 is dilute, $\gamma_1 \approx 1$ (close to pure 1)

then $-\ln x_1 = -\ln(1-x_2) \approx x_2 \approx \dfrac{\Delta \underline{H}_1^{fus}}{RT_f^{\,2}} \Delta T$ (a familiar result?)

binary mixture

$x_2 << 1$

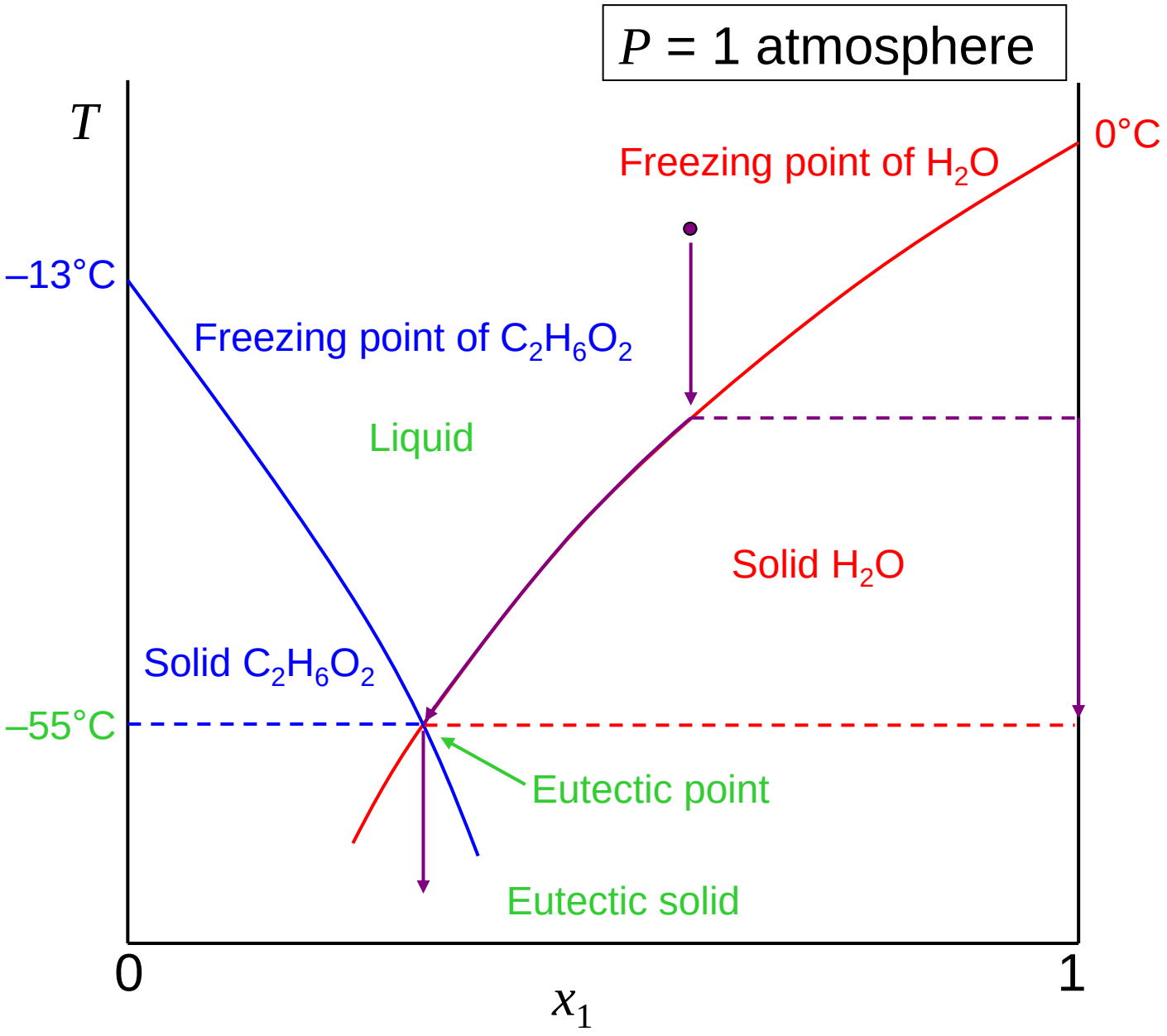
$\Delta \underline{H}_1^{fus}$ positive: heat in to melt

$\Rightarrow$ depression of freezing point

Example – freezing of an ethylene glycol, water liquid mixture

Note – depression of freezing point may be caused by two liquids mixing

$$1 = H_2O$$
$$2 = C_2H_6O_2$$


P = 1 atmosphere
T
0°C
Freezing point of H2O
–13°C
Freezing point of C2H6O2
Liquid
Solid H2O
Solid C2H6O2
–55°C
Eutectic point
Eutectic solid
0
1
x1

Assumption is that only pure solid freezes out – this is not always true, e.g. water of crystallisation

At eutectic – freezing point of 1 = freezing point of 2

Tiny domains of pure water or pure ethylene glycol freeze out simultaneously forming a eutectic solid. On a molecular scale solid phases are still pure

Activity Coefficient Models in Common Use

Random mixing models

e.g. Margules models

Random mixing implies we expect $\underline{S}^{ex} \approx 0$ and $\underline{G}^{ex} \approx \underline{H}^{ex}$

Parameters used in random mixing models have **no physical significance**

Very unsafe to extrapolate beyond the range of temperature and composition for which data were fit

This is generally true – even more so when models lack any physical basis

Redlich-Kister expansion:

zero when pure

$$\underline{G}^{ex} = x_1 x_2 [A + B(x_1 - x_2) + C(x_1 - x_2)^2 + \cdots]$$

polynomial expansion

One or two constants = Margules

Mathematically simple, as many parameters as desired

Good for hand or Excel calculations

Van Laar equation

The first activity coefficient model with adjustable parameters having physical significance – not better than the Redlich-Kister equation though

The van Laar equations were derived on the basis of the van der Waals EOS (van der Waals was van Laar's PhD supervisor!).

For a binary mixture, the equations are:

$$\ln\gamma_1 = \frac{\alpha}{\left(1+\frac{\alpha x_1}{\beta x_2}\right)^2} \qquad \ln\gamma_2 = \frac{\beta}{\left(1+\frac{\beta x_2}{\alpha x_1}\right)^2}$$

α and β are the model parameters

where $\alpha = \frac{b_1}{RT}\left(\frac{\sqrt{a_1}}{b_1} - \frac{\sqrt{a_2}}{b_2}\right)^2$ and $\beta = \frac{b_2}{b_1}\alpha$

a_1 a_2 b_1 b_2 being the parameters from the van der Waals EOS

Regular solution theory

$\underline{G}^{ex} = \underline{H}^{ex}$ and $\underline{V}^{ex} = \underline{S}^{ex} = 0$

random mixing assumption

$$\ln \gamma_1 = \frac{\underline{V}_1}{RT} \Phi_2{}^2 \left[(\delta_1 - \delta_2)^2 + 2 l_{12} \delta_1 \delta_2 \right]$$

$$\Phi_i = \frac{x_i \underline{V}_i}{\sum x_j \underline{V}_j}$$

l_{12} is an interaction parameter
i.e. a function of the nature of both 1 and 2
- usually ignored

$$\delta_i = \sqrt{\frac{\Delta \underline{H}_i^{VAP} - RT}{\underline{V}_i}}$$

solubility parameter
– a <u>pure</u> substance property

δ_is are tabulated for many compounds
– particularly solvents and polymers

RST is <u>very</u> useful for quick calculations – if l_{12} ignored

Flory-Huggins Model

for cases where one species is much larger than the other e.g. polymers

$$\Delta \underline{S}^{mix} = -R\left(x_1 \ln \phi_1 + x_2 \ln \phi_2 \right)$$

$$\phi_1 = \frac{x_1 \underline{V}_1}{x_1 \underline{V}_1 + x_2 \underline{V}_2} = \frac{x_1}{x_1 + m x_2} \quad \text{and} \quad \phi_2 = \frac{m x_2}{x_1 + m x_2} \quad \text{with} \quad m = \frac{\underline{V}_2}{\underline{V}_1}$$

Note: this is a random mixing model in volume fraction rather than mole fraction and so $\underline{S}^{ex} \neq 0$

and $$\Delta \underline{H}^{mix} = \underline{H}^{ex} = \chi RT \left(x_1 + m x_2 \right) \phi_1 \phi_2$$

χ is the variable parameter in this model

Note: analogy with one constant Margules model

Overall:

$$\frac{\underline{G}^{ex}}{RT} = \left(x_1 \ln \frac{\phi_1}{x_1} + x_2 \ln \frac{\phi_2}{x_2} \right) + \chi \left(x_1 + m x_2 \right) \phi_1 \phi_2$$

$\underline{S}^{ex}$ $\underline{H}^{ex}$

χ can be obtained from solubility parameters as for RST